Reg. No. | | | | | | | | | | | [14CE55]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2018

HYDROLOGY

Time:3 Hours Maximum Marks:70

Instructions: i. Answer any **FIVE** full Questions choosing at least two from each part.
ii. Missing data may be suitably assumed.

PART-A

1. a. Explain the hydrological cycle with the help of a neat sketch. 06

b. A catchment area of 150 Km^2 received 120cm of rainfall in a year. At the outlet of the catchment, the flow in the stream draining the catchment was found to have an average rate of (i) 1.5 m^3/s for the first 3 months, (ii) 2.0 m^3/s for the remaining 6 months, (iii) 3.5 m^3/s for the remaining 3 months. (a) What is the runoff co-efficient of the catchment? (b) If the afforestation of the catchment reduced the runoff coefficient to 0.35, what is the increase in abstraction from precipitation due to infiltration, evaporation and transpiration for the same annual rainfall of 120 cm? 08

2. a. Define rain gauge. Describe with a neat sketch the principle of working of Symon's Non-recording rain-gauge and its demerits. 07

b. Determine the additional number of rain gauges in a catchment from the following data: 07

i. No of existing rain gauge = 05

ii. Mean annual rainfall at the gauges are = 78.8cm, 90.2cm, 98.6cm, 102.4cm, and 70.4cm

iii. Estimated error = 6%

3. a. What is evapotranspiration? What are the factors that affect evapotranspiration? How would you measure evapotranspiration? 08

b. During a daily routine observation, 11 liters of water was added to bring the water surface in the evaporation pan to the stipulated level and nearby rain gauge measured 4mm of rainfall. What was the evaporation recorded for the day if the diameter of the pan is 125cm? 06

4. a. Explain with a neat sketch, the measurement of infiltration by double ring infiltrometer. 06

b. The rainfall rates for successive 30 minutes intervals upto 4 hours are given below. If the surface runoff is 3.6cm, determine ϕ and w indices. 08

Time (mins)	0	30	60	90	120	150	180	210	240
Rainfall Intensity (cm/h)	0	1.3	2.8	4.1	3.9	2.8	2.0	1.8	0.9

PART-B

5. a. List and explain the different components of runoff. **06**

   b. Explain different methods to establish rainfall – runoff relationship. **08**

6. a. Explain various methods of baseflow separation. **06**

   b. The hourly ordinates of a 2-hours unit hydrograph are given below. Derive a 6-hours unit hydrograph for the same catchment. **08**

Time (hours)	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Discharge (m^3/s)	0	1	2.7	5	8	9.8	9	7.5	6.3	5	4.0	2.9	2.1	1.3	0.5	0

7. a. Mention any two empirical formulae for estimating flood. **06**

   b. What are the different methods of flood control? Explain any two of them. **08**

8. a. Define all the Aquifer parameters. **06**

   b. List out the assumptions made in Dupuits's theory. **02**

   c. A well of 0.5 m diameter penetrates fully into a confined aquifer of thickness 20m and hydraulic conductivity 8.2×10^{-4} m/s. What is the discharge expected from this well if the drawdown in the well is not to exceed 3m. The radius of influence may be taken as 260m. **06**